# Qubits

Review Learning Objectives

**Assignment details**

**Due**
May 12, 11:59 PM IST

**Attempts**

Try again

**Submitted**
May 2, 12:00 AM IST

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**1.** Consider a qubit of the form **3 / 3 points**

$|\varphi\rangle = -\frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$

Select all the correct options about $|\varphi\rangle$ from the choices given below.

- [ ] The quantum state $|\varphi\rangle$ is identical to $|\xi\rangle = |0\rangle + i|1\rangle$. In other words, the factor $\frac{1}{\sqrt{2}}$ is not needed and serves just to scare beginners away from this area.
- [x] Measuring it yields $0$ and $1$ with equal probabilities $(\frac{1}{2})$

  **Correct**
  Correct.
- [ ] The quantum state $|\varphi\rangle$ is identical to $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$ since both states cannot be distinguished by measurements.
- [ ] The inner product $\langle 0|\varphi\rangle$ equals $\frac{1}{2}$.
- [x] Upon measurement the super position collapses to one of the pure states $|0\rangle$ or $|1\rangle$.

  **Correct**
  True

**2.** Which of the quantum states yields a probability of $\frac{3}{4}$ of yielding $|0\rangle$ upon measurement? **3 / 3 points**

$$ \frac{\sqrt{3}}{2} | 0 \rangle + \frac{1}{2} | 1 \rangle $$

What is the value of the inner product of two basis states $|0\rangle$ and $|1\rangle$, i.e, $\langle 0|1\rangle$? 0

Suppose $|\varphi\rangle$ is a qubit, what is the value of $\langle\varphi|\varphi\rangle$? 1

**Response 1**

- ( ) $\frac{\sqrt{3}}{2}|1\rangle + \frac{1}{2}|0\rangle$
- ( ) $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- (•) $\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$

**Correct**

**Response 2**

- ( ) -1
- (•) 0
- ( ) $i$
- ( ) 1

**Correct**
Correct: the pure states are orthogonal. Hence inner product is 0.

**Response 3**

- ( ) -1
- ( ) Varies, depending on the superposition.
- ( ) 0
- (•) 1

**Correct**

**3.** Select the correct value of the qubit $|-\rangle$ that is orthogonal to **1 / 1 point**

$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

In other words, $\langle +|-\rangle = 0$.

- ( ) $|-\rangle = |0\rangle$
- (•) $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- ( ) $|-\rangle = \frac{1}{4}(\sqrt{11}|0\rangle + \sqrt{5}|1\rangle)$
- ( ) $|-\rangle = \frac{1}{2}(\sqrt{3}|0\rangle - |1\rangle)$

**Correct**

Correct: $\langle -|+\rangle = \frac{1}{2}(\langle 0|0\rangle + \langle 0|1\rangle - \langle 1|0\rangle - \langle 1|1\rangle) = \frac{1}{2}(1+0+0-1) = 0.$